

* Suppose we receive a message w/ a string
of a 's & b 's: $aababbaaaabbb... \equiv \underline{X}$

Suppose a occurs w/ probability p and b w/ $\underline{1-p}$

Q: How many bits of information can we extract from a long message of this kind w/ N detectors

→ For large N , a occurs pN times while b occurs $(1-p)N$ times

- The number of possible messages:

$$\begin{aligned} (N! / (pN)! ((1-p)N)!) &\sim N^N / (pN)^{pN} ((1-p)N)^{(1-p)N} \\ &= 1 / (p^p (1-p)^{(1-p)})^N =: 2^{NS} \end{aligned}$$

where, S is the Shannon entropy

$$S := -p \log(p) - (1-p) \log(1-p)$$

$$= - \sum_i p_i \log(p_i)$$

where we have used the Stirling formula:

* Intuition:

We can interpret the Shannon entropy as either

- (i) Our uncertainty in X before we learn the value of X or,

Let's assume that much is no longer asked

a measure of how much info we have gained
after we learn the value of X

To each character we associate p_i
so we have a prob. dist. of the form
 $\underline{X} = (p_1, \dots, p_n)$ We thus define:

$$S(\underline{X}) := S(p_1, \dots, p_n) := - \sum_i p_i \log(p_i)$$

(Shannon Entropy)

* We take $\log(\cdot) \equiv \log_2(\cdot)$

• Note that terms for $p_i = 0$ are defined
since
$$\lim_{x \rightarrow 0} (x \log(x)) = \lim_{x \rightarrow 0} \left(\frac{\log(x)}{1/x} \right) \sim \frac{-\infty}{\infty}$$
$$\text{L'Hop: } \lim_{x \rightarrow 0} \left(\frac{1/x}{-1/x^2} \right) = \lim_{x \rightarrow 0} -x \rightarrow 0 \quad \checkmark$$

* This def. also allows us to quantify the
resources needed to store information.

Suppose we have a radio antenna (i.e. a
Source)



$X_1 X_2 \dots$

• Each are independent, identically
distributed.

Q: What minimal physical resources are required to store the info being produced by the source.

A: $S(X)$ bits are required per source symbol
Where $S(X) = S(x_1) = S(x_2) = \dots$ is the entropy of each random variable modeling the source.

Shannon's noiseless coding theorem

ex: Suppose an information source produces one of four symbols: x_1, x_2, x_3, x_4

- Without compression two bits of storage space corresponding to the four possible outputs are consumed for each use of the source.

$$p_1 = \frac{1}{2}, p_2 = \frac{1}{4}, p_3 = \frac{1}{8}, p_4 = \frac{1}{8}$$

→ A data compression scheme:

$$x_1: 0, x_2: 10, x_3: 110, x_4: 111$$

$$\text{average length of the compressed string} \\ \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = \frac{7}{4} \text{ bits}$$

of info per source.

Which is less than the 'naïve' approach of storing the source.

⇒ This matches the Entropy

$$S(X) = -1/2 \log(1/2) - 1/4 \log(1/4) - 1/4 \log(1/8) \\ = \frac{2}{4} + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 3 = 7/4!$$

- Any attempts to compress the source further results in data being irretrievably lost.

* Fundamental measures of information arises as the answers to fundamental questions about the physical resources required to solve some information processing problem.

→ Intuitive justification of $H(X)$

Define the information fⁿ: $I(E)$, whose value is determined by an event E .

(i) $I(E)$ is a fⁿ only of the prob. of the event $E \Rightarrow I = I(p)$

(ii) $I \in C^\infty(\mathbb{R})$

(iii) $I(pq) = I(p) + I(q)$ when $p, q > 0$

(The information gained when two ind. events occur with individual prob p, q is the sum of info gained from each source.)

Claim: $I(p) = k \log(p)$, $k \in \mathbb{R}$

Let $I(p) = \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!} p^n$

$$\Rightarrow n=1: p+q = pq \Rightarrow p = \frac{q}{(q-1)}$$

$$\begin{aligned} \therefore I(p, q) &= \sum_{n=1}^{\infty} \frac{f(n, q)}{n!} \left(\frac{q^2}{q-1} \right)^n \\ &= \sum_{n=1}^{\infty} \frac{f(n, q)}{n!} (q^2 (1 - q + \dots))^n \\ &= \dots \end{aligned}$$

Not sure you can obtain it this way
just simply assume $I(p) = k \log(p)$

$$\therefore k(\log(pq)) = k \log(p) + k \log(q)$$

$$\Rightarrow I(p_i) = k \log(p_i)$$

The average info gain when one of mutually exclusive set of events w/ prob. $p_1 \dots p_n$ occurs

$$is \quad p_1 I(p_1) + \dots = \sum_i p_i I(p_i)$$

$$S = k \sum_i p_i I(p_i)$$

Basic Properties of Entropy:

The Binary entropy:

$$S_{bin}(p) := -p \log(p) - (1-p) \log(1-p)$$

If we can plot this:

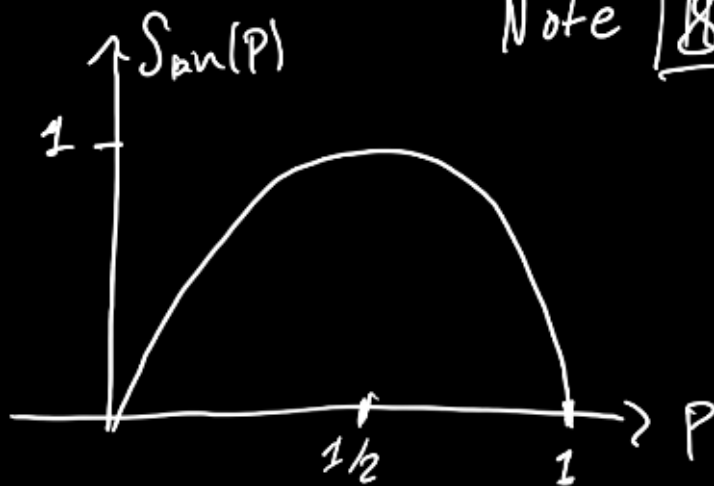
$$\begin{aligned} \frac{d S_{bin}}{d p} &= -\log(p) - 1 + \log(1-p) + 1 = 0 \\ &= \log\left(\frac{1}{p} - 1\right) = 0 \end{aligned}$$

$$\begin{aligned} \therefore 1 &= \frac{1}{p} - 1 \Leftrightarrow \boxed{p = 1/2} \text{ Maxima} \\ \frac{d^2 S_{bin}}{d p^2} &= -\frac{1}{p} = \frac{1}{1-p} \\ \frac{d^2 S}{d p^2} \bigg|_{p=1/2} &< 0 : \text{local Maximum} \end{aligned}$$

$H_{\text{bin}}(p) \approx 0$: Easy to see both log terms
must vanish or its coeff.

$$p=0, p=1$$

$$\text{Note } \boxed{H(p) = H(1-p)}$$



*

the first of these is the fact that the majority of the population is now living in urban areas. This has led to a concentration of people in a few large cities, which has in turn led to a number of problems. One of the most serious is the lack of adequate housing. In many of these cities, the housing is of a very poor quality and is often overcrowded. This has led to a number of health problems, particularly in the case of children. Another problem is the lack of adequate sanitation. In many of these cities, there is no proper sewage system, and the waste is often dumped in the streets. This has led to a number of diseases, particularly in the case of children. A third problem is the lack of adequate education. In many of these cities, there are no schools, and the children are often left to fend for themselves. This has led to a number of social problems, particularly in the case of children.

The second of these is the fact that the majority of the population is now living in rural areas. This has led to a number of problems. One of the most serious is the lack of adequate housing. In many of these areas, the housing is of a very poor quality and is often overcrowded. This has led to a number of health problems, particularly in the case of children. Another problem is the lack of adequate sanitation. In many of these areas, there is no proper sewage system, and the waste is often dumped in the streets. This has led to a number of diseases, particularly in the case of children. A third problem is the lack of adequate education. In many of these areas, there are no schools, and the children are often left to fend for themselves. This has led to a number of social problems, particularly in the case of children.

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The sixth of these is the fact that the majority of the population is now living in inland areas. This has led to a number of problems. One of the most serious is the lack of adequate housing. In many of these areas, the housing is of a very poor quality and is often overcrowded. This has led to a number of health problems, particularly in the case of children. Another problem is the lack of adequate sanitation. In many of these areas, there is no proper sewage system, and the waste is often dumped in the streets. This has led to a number of diseases, particularly in the case of children. A third problem is the lack of adequate education. In many of these areas, there are no schools, and the children are often left to fend for themselves. This has led to a number of social problems, particularly in the case of children.

The relative Entropy:

- The relative entropy is a very useful entropy which measures the closeness of two prob. $p(x)$ and $q(x)$.
- Let $P_{X,Y}(x_i, y_i)$ be the prob. that, in a given occurrence, Alice sends $X = x_i$ and Bob hears $Y = y_i$.
The prob. that Bob hears $Y = y_i$, summing over all choices of what Alice intended is,
$$P(y_i) = \sum_j P(x_j, y_i)$$

$$P_Y(y_i) = \sum_j P_{X,Y}(x_j, y_i)$$

- If Bob does hear $Y = y_j$, his estimate of the prob. that Alice sent x_i is the conditional probability:

$$P_{X|Y}(x_i | y_j) := \frac{P_{X,Y}(x_i, y_j)}{P_Y(y_j)}$$

From Bob's point of view, once he has heard $Y = y_j$, his estimate of the remaining entropy in Alice's signal

$$S_{X|Y=y_j} = - \sum_i P_{X|Y}(x_i, y_j) \log(P_{X|Y}(x_i, y_j))$$

(Relative Entropy)

$$\begin{aligned} S(P(X) \| Q(X)) &= \sum_x P(X) \log\left(\frac{P(X)}{Q(X)}\right) \\ &= -S(X) - \sum_x P(X) \log(Q(X)) \end{aligned}$$

Polyakov symmetries:

We must distinguish between Spacetime on M^D and the WS symm. acting on Σ .

(i) Spacetime symmetries:

We have Poincaré invariance

$$X^\mu(\sigma) \mapsto \Lambda^\mu_\nu (X^\nu - a^\nu), \quad \Lambda \in SO(1, d+1)$$

Recall: $(M_{[pq]})^\mu_\nu := \xi^\mu_p \eta_{q\nu} - \delta^\mu_q \eta_{p\nu}$

$$\Leftrightarrow \delta X^\mu = \lambda^\mu_\nu X^\nu + a^\mu$$

$$\delta h_{\alpha\beta} = 0$$

(local) WS symmetries:

→ Reparametrization invariance

$$\sigma^a \mapsto \sigma'^a := \sigma^a - \xi^a(\sigma)$$

(i) Scalar: $x^\mu(\sigma) \mapsto x'^\mu(\sigma') = x^\mu(\sigma(\sigma'))$

$$x'^\mu(\sigma') = x^\mu(\sigma') + \xi^a \partial_a x^\mu|_\sigma$$

$$\Rightarrow \delta x^\mu = x'^\mu(\sigma) - x^\mu(\sigma) = \xi^a \partial_a x^\mu(\sigma)$$

(ii) Metric:

$$\delta h_{\alpha\beta} = \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha$$

(iii) Fun:

$$\delta \Gamma_h = \frac{1}{2} h^{\alpha\beta} (\xi^\gamma \partial_\gamma h_{\alpha\beta} + \partial_\alpha \xi^\gamma h_{\gamma\beta} + \partial_\beta \xi^\gamma h_{\alpha\gamma})$$

note $\partial_\alpha \Gamma_h = -\frac{1}{2} \frac{1}{\Gamma_h} h^{\gamma\delta} \partial_\alpha h_{\gamma\delta}$

$$\Leftrightarrow \delta \Gamma_h = \partial_\alpha (\xi^\alpha \Gamma_h)$$

* For $\Lambda^\mu_\nu = \delta^\mu_\nu + \lambda^\mu_\nu$

$$\Leftrightarrow \delta x^\mu = \lambda^\mu_\nu x^\nu$$

To see that λ is antisymmetric:

$$\eta_{\mu\nu} x^\mu x^\nu = \eta_{\mu\nu} x'^\mu x'^\nu$$

$$\Leftrightarrow \delta(\eta_{\mu\nu} x^\mu x^\nu) = 0$$

$$2 \eta_{\mu\nu} \lambda^\mu_\sigma x^\sigma x^\nu = 0$$

$$2 \underbrace{\lambda_{\nu\sigma}}_{\text{Antisym}} \underbrace{x^\sigma x^\nu}_{\text{Sym}} = 0$$

examples of Poincare invariance:

(i) Rotation about X^3 by an angle θ

$$x'^1 = \cos(\theta) x^1 + \sin(\theta) x^2$$

$$x'^2 = -\sin(\theta) x^1 + \cos(\theta) x^2$$

$$\theta \ll 1, \quad \cos \rightarrow 1, \quad \sin \rightarrow \theta$$

$$x'^1 = x^1 + \theta x^2$$

$$x'^2 = -\theta x^1 + x^2$$

$$\delta x^1 = \theta x^2, \quad \delta x^2 = -\theta x^1$$

$$\text{but since } \delta x^m = a^m_{\nu} x^{\nu}$$

$$a^1_2 = \theta, \quad a^2_1 = -\theta \quad (\Rightarrow) \quad a_{12} = \eta_{1\lambda} a^{\lambda}_2 = \theta$$

(ii) Boost in the x^0 & x^1 direction by φ .

$$x'^0 = \cosh(\varphi) x^0 + \sinh(\varphi) x^1$$

$$x'^1 = \sinh(\varphi) x^0 + \cosh(\varphi) x^1$$

$$\cosh \rightarrow 1, \quad \sinh \rightarrow \varphi$$

$$\delta x^0 = \varphi x^1, \quad \delta x^1 = \varphi x^0$$

$$\Rightarrow a^0_1 = \varphi, \quad a^1_0 = \varphi$$